

The reaction of Compound **1** with HBr results in an electrophilic addition across a double bond that proceeds in two steps, as shown in Figure 1.

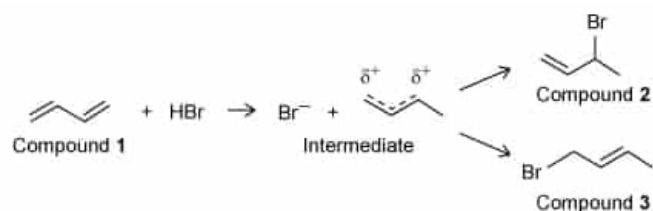


Figure 1 Electrophilic addition of HBr to 1,3-butadiene (Compound **1**)

In the first step, the electron-rich pi bond of Compound **1** forms a new bond with the electron-deficient hydrogen atom donated by HBr. This results in the formation of a resonance-stabilized carbocation intermediate. In the second step, the electron-deficient carbocation intermediate reacts with the bromide nucleophile to form either Compound **2** or Compound **3**. At lower reaction temperatures, Compound **2** is the major product, but at higher temperatures, Compound **3** is the major product. A diagram showing the energy profile for the reaction is given in Figure 2.

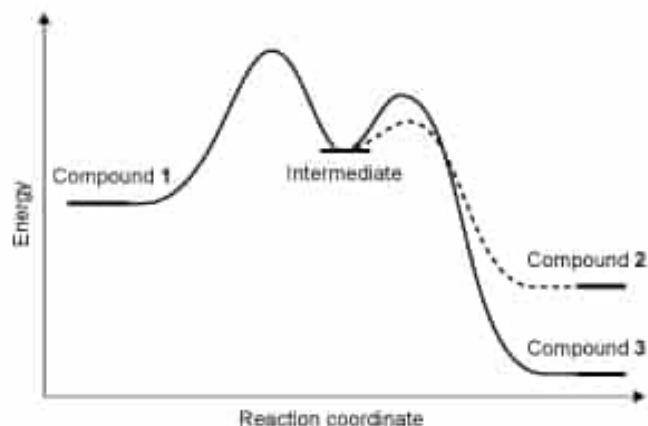
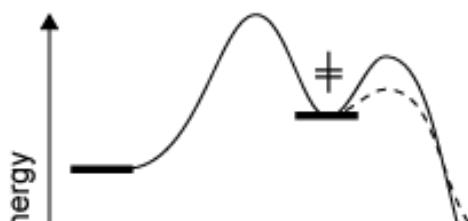
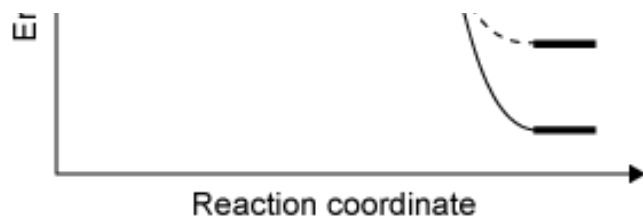


Figure 2 Energy profile for the electrophilic addition of HBr to 1,3-butadiene (Compound **1**)

Which of the following diagrams correctly labels the position of the transition state ($\ddagger$) for the first step of the reaction shown in Figure 1?

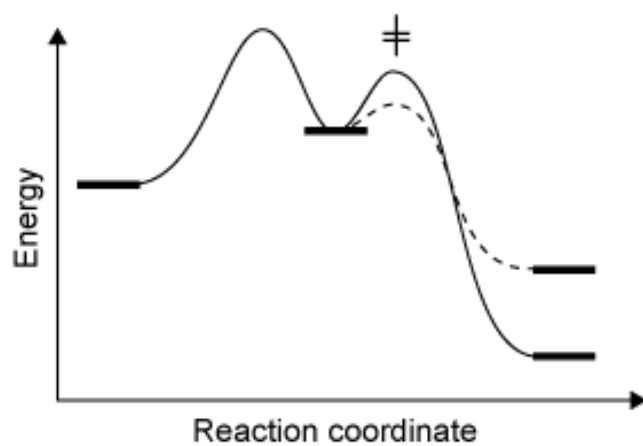
☐ A.





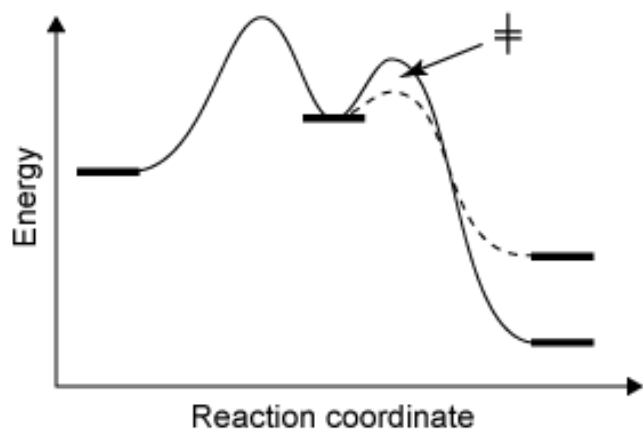
[21%]

☐ B.



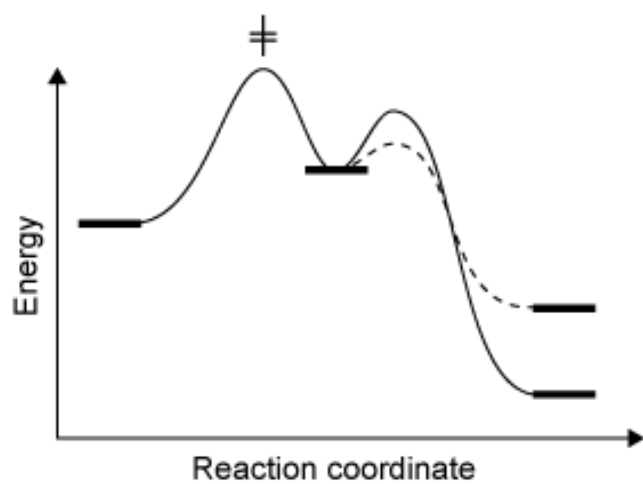
[2%]

☐ C.



[2%]

☒ D.

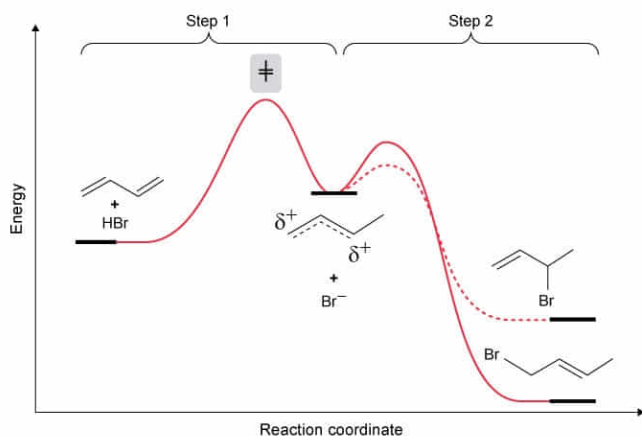


[73%]

Correct

72% answered correctly

Explanation:



A **reaction free-energy diagram** is a plot of the **Gibbs free energy** (y-axis) versus the progress of a reaction pathway. The reaction progress is represented abstractly by the **reaction coordinate** (x-axis), which is a variable that correlates to changes in molecular geometry as a reactant molecule is converted into a product molecule.

During a given step in a molecular conversion, existing bonds are broken while new bonds are formed. This results in a momentary **transition state** ($\ddagger$) (activated complex) with elevated energy caused by increased geometric strain and charge repulsion. As a result, transition states are seen graphically as higher **energy maxima** (peaks). Reactions involving more than one elementary step will exhibit more than one peak, indicating a separate transition state for each elementary step.

The free-energy diagram for the electrophilic addition of HBr to Compound **1** shown in the passage has two peaks. Each peak corresponds to the transition state of a separate elementary reaction step. Because peaks farther along the reaction coordinate axis occur *after* preceding peaks, the peak closest to the reactants is the **transition state** for the first step of the reaction.

(Choice A) This labeled location of the plot is a lower energy minimum between two transition states. This is the intermediate, not a transition state.

(Choices B and C) These peaks are **transition states** associated with the formation of Compounds **2** and **3**, but these peaks are farther along the reaction coordinate axis and correspond to the *second step* of the reaction.

Educational objective:

A reaction free-energy diagram is a plot of the Gibbs free energy (y-axis)

versus the reaction coordinate (x-axis). During elementary molecular conversion steps, existing bonds are broken while new bonds are formed, resulting in momentary transition states with elevated energy that are seen graphically as energy maxima (peaks).

Time spent:0 sec

Subject:
General Chemistry

QID:401712

System:
Thermodynamics, Kinetics & Gas Laws

The reaction of Compound **1** with HBr results in an electrophilic addition across a double bond that proceeds in two steps, as shown in Figure 1.

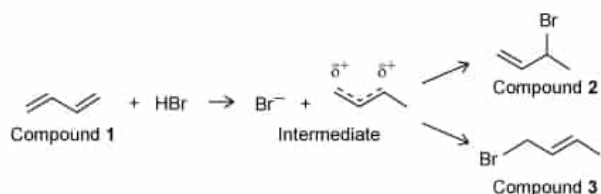


Figure 1 Electrophilic addition of HBr to 1,3-butadiene (Compound **1**)

In the first step, the electron-rich pi bond of Compound **1** forms a new bond with the electron-deficient hydrogen atom donated by HBr. This results in the formation of a resonance-stabilized carbocation intermediate. In the second step, the electron-deficient carbocation intermediate reacts with the bromide nucleophile to form either Compound **2** or Compound **3**. At lower reaction temperatures, Compound **2** is the major product, but at higher temperatures, Compound **3** is the major product. A diagram showing the energy profile for the reaction is given in Figure 2.

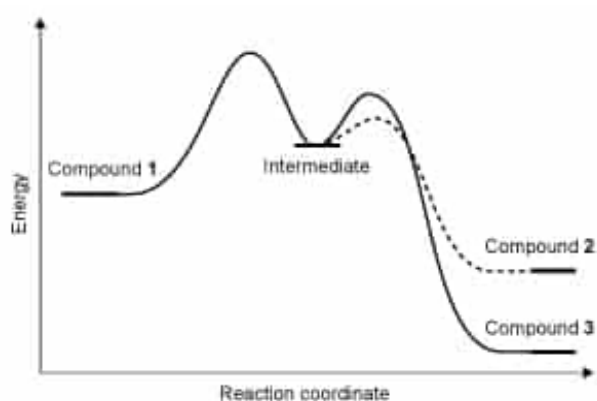
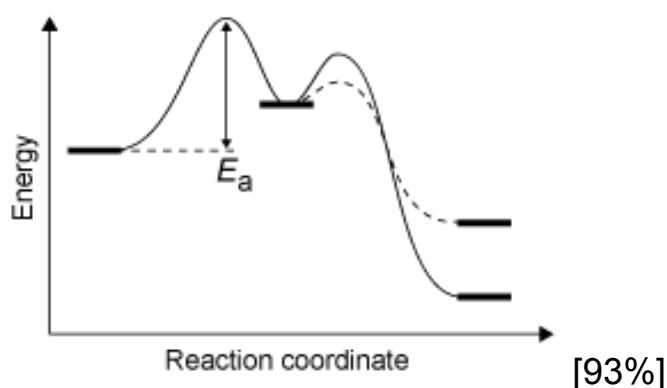


Figure 2 Energy profile for the electrophilic addition of HBr to 1,3-butadiene (Compound **1**)

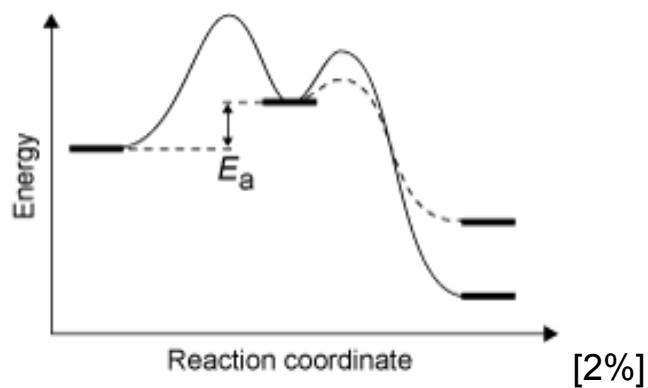
Which of the following diagrams correctly labels the activation energy E_a for the reaction shown in Figure 1?

✓ ☒ A.

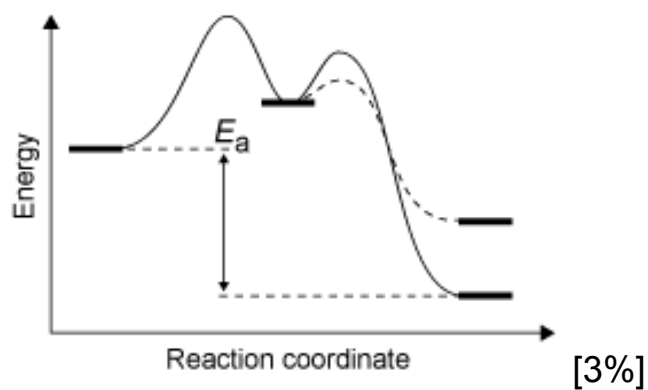


[93%]

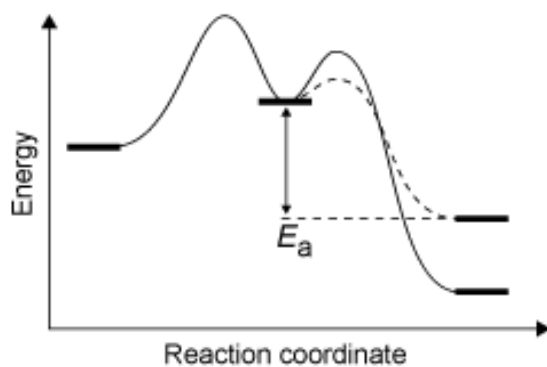
☐ B.



☐ C.



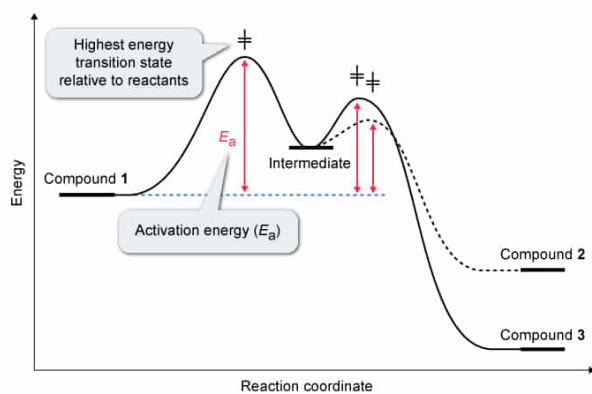
☐ D.



Correct

93% answered correctly

Explanation:



As existing bonds are broken and new bonds are formed for the conversion of a

reactant to a product, the elevated energy necessary to form a transition state acts as an energy barrier that must be overcome for the reaction to proceed. The transition state ($\ddagger$) with the highest energy (relative to the reactants) presents an energy barrier to the overall reaction. Therefore, the energy necessary to achieve the highest energy transition state is the **minimum energy** needed to activate the reactants sufficiently for the **completion of the reaction** (the activation energy).

In a reaction free-energy diagram, transition states are seen graphically as energy maxima (peaks). The **highest energy transition state** (tallest peak **relative to the reactant** energy) is the greatest energy barrier for the overall reaction and determines the **activation energy** E_a . Therefore, E_a is the energy difference between the highest peak and the reactants (peak energy minus reactant energy).

(Choice B) This is the energy difference between the intermediate and the reactants. The activation energy is larger because the highest energy transition state must be reached to form the intermediate of this reaction.

(Choice C) This is the overall change in energy for the reaction to form Compound 3. This does not reflect the energy required to achieve the highest energy transition state along the reaction pathway.

(Choice D) This is the energy difference between the intermediate and Compound 2. This shows the relative change in energy between the species, but it does not reflect the activation energy associated with the highest energy transition state.

Educational objective:

The activation energy is the minimum energy needed to reach the highest energy transition state that enables the completion of a reaction. In a reaction free-energy diagram, the activation energy is the energy difference between the tallest peak (relative to the reactants) and the reactants (peak energy minus reactant energy).

Time spent: 0 sec

Subject:
General Chemistry

QID: 401713

System:
Thermodynamics, Kinetics & Gas Laws

The reaction of Compound 1 with HBr results in an electrophilic addition across a double bond that proceeds in two steps, as shown in Figure 1.

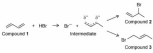


Figure 1 Electrophilic addition of HBr to 1,3-butadiene (Compound 1)

In the first step, the electron-rich pi bond of Compound 1 forms a new bond with the electron-deficient hydrogen atom donated by HBr. This results in the formation of a resonance-stabilized carbocation intermediate. In the second step, the electron-deficient carbocation intermediate reacts with the bromide nucleophile to form either Compound 2 or Compound 3. At lower reaction temperatures, Compound 2 is the major product, but at higher temperatures, Compound 3 is the major product. A diagram showing the energy profile for the reaction is given in Figure 2.

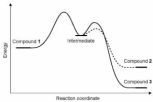


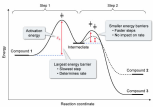
Figure 2 Energy profile for the electrophilic addition of HBr to 1,3-butadiene (Compound 1)

The rate-determining step for the overall reaction in Figure 1 is the conversion of:

- ☒ A. Compound 1 into the intermediate. [91%]
☐ B. Compound 1 into Compound 3. [1%]
☐ C. the intermediate into Compound 2. [3%]
☐ D. the intermediate into Compound 3. [3%]

Correct
91% answered correctly

Explanation:



For a general reaction in which x moles of reactant B and y moles of reactant C form z moles of product D, the reaction equation is



The rate of this reaction can be expressed by the general rate law:

$$\text{Rate} = k [\text{B}]^m [\text{C}]^n$$

where k is the temperature-dependent rate coefficient, $[\text{B}]$ and $[\text{C}]$ are the molar concentrations of the reactants, and m and n are the partial reaction orders (determined experimentally). The temperature dependence of k is expressed by the Arrhenius equation:

$$k = A e^{-\left(\frac{E_a}{RT}\right)}$$

The rate law shows that the reaction rate is proportional to k , and the Arrhenius equation shows that k decreases exponentially as the activation energy E_a increases. If the reaction occurs in two steps, the rate-determining step of the reaction will be the step with the largest energy barrier (the slowest step).

The free energy diagram (Figure 2) of the reaction in Figure 1 shows that forming either product involves two steps. The energy barrier for the conversion of Compound 1 into the intermediate (Step 1) is much larger than either of the energy barriers to convert the intermediate into either product (Step 2). Therefore, the conversion of Compound 1 into the intermediate (Step 1) is the rate-determining step (slowest step with the smallest value of k) and determines the activation energy for the reaction.

(Choice B) The conversion of Compound 1 into Compound 3 involves two steps. Because Step 2 is much faster, only Step 1 (the slowest step) is the overall rate-determining step.

(Choices C and D) The energy barrier (height of the peak) relative to the energy of the intermediate (Step 2) is much smaller than the energy barrier between the intermediate and Compound 1. Larger energy barriers have slower rates and yield smaller values of k .

Educational objective:

The reaction rate is proportional to the rate coefficient k , which decreases exponentially as the energy needed to reach the transition state increases. In a reaction involving two or more steps, the step that must overcome the largest energy barrier to pass through the transition state is the rate-determining step.

Time spent: 2 sec
Subject:
General Chemistry

QID: 401714
System:
Thermodynamics, Kinetics & Gas Laws

The reaction of Compound 1 with HBr results in an electrophilic addition across a double bond that proceeds in two steps, as shown in Figure 1.

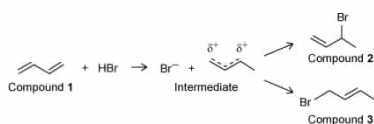


Figure 1 Electrophilic addition of HBr to 1,3-butadiene (Compound 1)

In the first step, the electron-rich pi bond of Compound 1 forms a new bond with the electron-deficient hydrogen atom donated by HBr. This results in the formation of a resonance-stabilized carbocation intermediate. In the second step, the electron-deficient carbocation intermediate reacts with the bromide nucleophile to form either Compound 2 or Compound 3. At lower reaction temperatures, Compound 2 is the major product, but at higher temperatures, Compound 3 is the major product. A diagram showing the energy profile for the reaction is given in Figure 2.

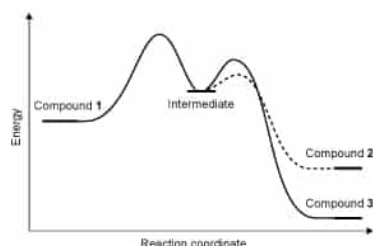


Figure 2 Energy profile for the electrophilic addition of HBr to 1,3-butadiene (Compound 1)

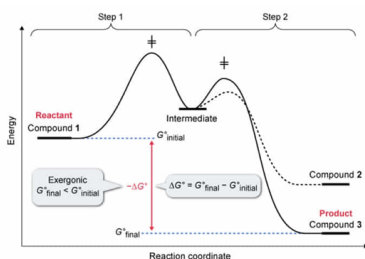
The reaction responsible for the conversion of Compound 1 into Compound 3 is:

- ☐ A. endergonic because the energy of the intermediate is greater than the energy of Compound 1. [3%]
- ☐ B. endergonic because the energy of the transition state is greater than the energy of Compound 1. [3%]
- ☐ C. exergonic because the energy of Compound 2 is greater than the energy of Compound 3. [2%]
- ☒ D. exergonic because the energy of Compound 3 is less than the energy of Compound 1. [90%]

Correct

90% answered correctly

Explanation:



The change in Gibbs free energy (ΔG) that occurs during a reaction process is evaluated by the difference in energy between the final state (products) and the initial state (reactants), which is expressed mathematically as:

$$\Delta G = (\text{Gibbs free energy})_{\text{products}} - (\text{Gibbs free energy})_{\text{reactants}}$$

If the energy of the products is less than the energy of the reactants, a **negative ΔG** will be obtained, indicating that the reaction is **exergonic** (energy is released from the reaction). Conversely, if the energy of the products is greater than the energy of the reactants, a **positive ΔG** will be obtained, indicating that the reaction is **endergonic** (energy is consumed by the reaction).

The Gibbs free energy diagram (Figure 2) for the conversion of Compound 1 into Compound 3 (Figure 1) shows that the energy of the product (Compound 3) is less than the energy of the reactant (Compound 1). As a result, ΔG for this conversion is negative, which indicates that the reaction is exergonic.

(Choices A and B) The fact that the transition state and the intermediate are higher in energy than Compound 1 makes the *first step* of the reaction endergonic, but the *overall reaction* is exergonic because the product formed in the second step (Compound 3) has a lower

energy than the reactant (Compound 1).

(Choice C) Energy changes associated with the formation of a given product in a reaction must be evaluated with respect to the energy of the reactants, not to the energy of another product.

Educational objective:

The change in Gibbs free energy (ΔG) during a reaction is evaluated as $\Delta G = (\text{Gibbs free energy})_{\text{products}} - (\text{Gibbs free energy})_{\text{reactants}}$. If the energy of the products is less than the energy of the reactants ($-\Delta G$), the reaction is exergonic (energy is released), but in the opposite case ($+\Delta G$), the reaction is endergonic (energy is consumed).

Time spent: 0 sec

Subject:
General Chemistry

QID: 401715

System:
Thermodynamics, Kinetics & Gas Laws

The reaction of Compound 1 with HBr results in an electrophilic addition across a double bond that proceeds in two steps, as shown in Figure 1.

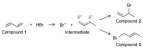


Figure 1 Electrophilic addition of HBr to 1,3-butadiene (Compound 1)

In the first step, the electron-rich pi bond of Compound 1 forms a new bond with the electron-deficient hydrogen atom donated by HBr. This results in the formation of a resonance-stabilized carbocation intermediate. In the second step, the electron-deficient carbocation intermediate reacts with the bromide nucleophile to form either Compound 2 or Compound 3. At lower reaction temperatures, Compound 2 is the major product, but at higher temperatures, Compound 3 is the major product. A diagram showing the energy profile for the reaction is given in Figure 2.

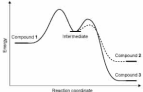


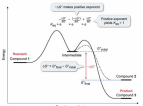
Figure 2 Energy profile for the electrophilic addition of HBr to 1,3-butadiene (Compound 1)

Which expression correctly describes the equilibrium constant K_{eq} for the formation of Compound 2 from the intermediate?

- ☐ A. $K_{eq} < 1$ [23%]
- ☐ B. $K_{eq} = 0$ [1%]
- ☒ C. $K_{eq} > 1$ [72%]
- ☐ D. $K_{eq} = \Delta G^\circ$ [2%]

Correct
72% answered correctly

Explanation:



For a general reaction in which x moles of reactant B and y moles of reactant C form z moles of product D, the reaction equation is



Using the molar concentrations of each species [B], [C], and [D] each raised to the power of its respective reaction coefficient, the **equilibrium constant** K_{eq} for the reaction is expressed as a ratio of the products over the reactants:

$$K_{eq} = \frac{[D]^z}{[B]^x [C]^y}$$

If the equilibrium favors the formation of products, then $[D]^z > [B]^x [C]^y$ and $K_{eq} > 1$. Conversely, if the formation of the products is unfavorable, then $[D]^z < [B]^x [C]^y$ and $K_{eq} < 1$.

The relationship between the **Gibbs free energy** ΔG° of a reaction and K_{eq} is given by the relationship:

$$\Delta G^\circ = -RT \ln K_{eq}$$

where R is the gas constant and T is the absolute temperature. Solving this equation for K_{eq} gives

$$K_{eq} = e^{-\frac{\Delta G^\circ}{RT}}$$

This relationship shows that when ΔG° is **negative**, the exponent term will become positive and make $K_{eq} > 1$, indicating that the equilibrium **favors product** formation. Because Figure 2 shows that ΔG is negative for the formation of Compound 2 from the intermediate, $K_{eq} > 1$ for the conversion.

(Choice A) If $K_{eq} < 1$, the equilibrium would be shifted toward the reactants and indicate a reaction with a positive ΔG . The formation of Compound 2 from the intermediate has a negative ΔG° .

(Choice B) $K_{eq} = 0$ indicates that absolutely no product is formed. If a chemical reaction takes place, K_{eq} must be a nonzero value.

(Choice D) ΔG° is proportional to the natural log of K_{eq} . ΔG° and K_{eq} cannot be equal at temperatures that permit the reaction to occur.

Educational objective:

The Gibbs free energy ΔG° and the equilibrium constant K_{eq} of a reaction are related through the gas constant R and absolute temperature T by the equation $\Delta G^\circ = -RT \ln K_{eq}$. Negative values of ΔG° indicate that $K_{eq} > 1$ and that the equilibrium favors product formation whereas positive values of ΔG° indicate that $K_{eq} < 1$ with an equilibrium that favors the reactants.

Time spent: 0 sec
Subject:
General Chemistry

QID:401716
System:
Thermodynamics, Kinetics & Gas Laws

The reaction of Compound 1 with HBr results in an electrophilic addition across a double bond that proceeds in two steps, as shown in Figure 1.

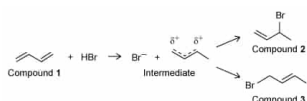


Figure 1 Electrophilic addition of HBr to 1,3-butadiene (Compound 1)

In the first step, the electron-rich pi bond of Compound 1 forms a new bond with the electron-deficient hydrogen atom donated by HBr. This results in the formation of a resonance-stabilized carbocation intermediate. In the second step, the electron-deficient carbocation intermediate reacts with the bromide nucleophile to form either Compound 2 or Compound 3. At lower reaction temperatures, Compound 2 is the major product, but at higher temperatures, Compound 3 is the major product. A diagram showing the energy profile for the reaction is given in Figure 2.

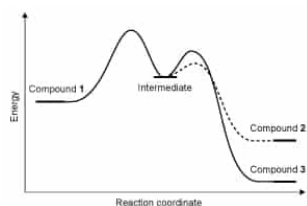


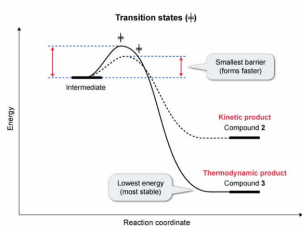
Figure 2 Energy profile for the electrophilic addition of HBr to 1,3-butadiene (Compound 1)

Based on the reaction in Figure 1 and the energy profile in Figure 2, which statement correctly describes both Compound 2 and Compound 3?

- ☐ A. Compound 2 and Compound 3 are both the kinetic products of the reaction. [2%]
- ☒ B. Compound 2 is the kinetic product, and Compound 3 is the thermodynamic product. [74%]
- ☐ C. Compound 2 is the thermodynamic product, and Compound 3 is the kinetic product. [14%]
- ☐ D. Compound 2 and Compound 3 are both the thermodynamic products of the reaction. [8%]

Correct
75% answered correctly

Explanation:



When a reaction can produce more than one product, how much of each product is formed often depends on thermodynamic and kinetic considerations. During product formation, the compound with the **lowest energy transition state** (lowest energy barrier for formation) **forms fastest**, making it the **kinetic product**. The compound with the **most stable** (lowest energy) structure is the **thermodynamic product**, but because it has a higher energy transition state (higher energy barrier for formation), it forms more slowly.

Because the kinetic product has a lower energy barrier for formation, at lower temperatures the kinetic product will be the major product produced. In such a case, the reaction is said to be under kinetic control. However, at higher temperatures the additional energy enables the reaction to overcome the larger energy barrier for the formation of the thermodynamic product, making it the major product formed. In this alternate case, the reaction is said to be under thermodynamic control.

For the reaction to form Compounds 2 and 3, the energy diagram of the reaction shows that Compound 2 has a lower energy transition state than Compound 3. Therefore, Compound 2 will form faster and is the kinetic product. Compound 3 has a higher energy transition state, but it is more stable (lower energy) than Compound 2. Therefore, Compound 3 is the thermodynamic product.

(Choice A) Compounds 2 and 3 have transition states of different energies and form at different rates. Both cannot be the kinetic product.

(Choice C) Compound 2 forms faster and is higher in energy than Compound 3. Therefore, Compound 2 is the kinetic product, not Compound 3.

(Choice D) Compounds 2 and 3 are not equally stable (they have different energies). Both cannot be the thermodynamic product.

Educational objective:

For a reaction producing more than one product, the product with the lowest energy transition state will form fastest and is the kinetic product. The compound with a higher energy transition state that produces the lowest energy (most stable) structure is the thermodynamic product.

Time spent: 0 sec
Subject:
General Chemistry

QID: 401717
System:
Thermodynamics, Kinetics & Gas Laws